

Advanced (Carolina Reaper)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
- (B) $ax + bx^2 + c$
- (C) $ax^2 - bx + c$
- (D) $ax + b$
- (E) $ax^2 + c$

Q2. Identify the quadratic function: $f(x) = 2x^2 - 5x + 1$

- (A) Linear function
- (B) Quadratic function in standard form
- (C) Cubic function
- (D) Exponential function
- (E) Rational function

Q3. A shopper spends 10 more on clothes each day. If she spent 20 on the first day, what is the quadratic equation for her total spending?

- (A) $y = x^2 + 10x + 20$
- (B) $y = 10x^2 + 20$
- (C) $y = x^2 + 20$
- (D) $y = 10x + 20$
- (E) $y = x^2 + 10x$

Q4. A photographer charges 50 for the first hour and 25 for each additional hour. What is the quadratic equation for the total cost?

- (A) $y = 25x^2 + 50$
- (B) $y = 25x + 50$
- (C) $y = x^2 + 25x + 50$
- (D) $y = 25x^2 - 50$
- (E) $y = x^2 + 50$

Q5. What is the key feature of the quadratic function $f(x) = -2x^2 + 3x - 1$?

- (A) Opens upwards
- (B) Opens downwards
- (C) Has a maximum value
- (D) Has a minimum value
- (E) Is a linear function

Q6. A quadratic function has a vertex at (2, 3). What is the equation of the axis of symmetry?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q7. A quadratic equation is in the form $ax^2 + bx + c$. What is the value of a in the equation $2x^2 - 3x + 1 = 0$?

- (A) 1
- (B) 2
- (C) 3
- (D) -2
- (E) -3

Q8. Identify the quadratic function $f(x) = x^2 - 4x + 4$

- (A) A parabola that opens upwards
- (B) A parabola that opens downwards
- (C) A horizontal line
- (D) A vertical line
- (E) A circle

Open Ended Questions (FRQ)

Q9. A water tank can hold 1000 gallons of water. Due to a leak, the tank is losing water at a rate of $2x^2 + 5x - 3$ gallons per hour, where x is the number of hours. Write an equation for the total amount of water in the tank after x hours.

Q10. A quadratic function has its vertex at $(-2, 1)$ and passes through the point $(1, 5)$. Find the equation of the quadratic function in standard form.

Chef's Tips

- Provide clear instructions for each question
- Ensure students have the necessary materials
- Encourage students to ask questions if they are unsure